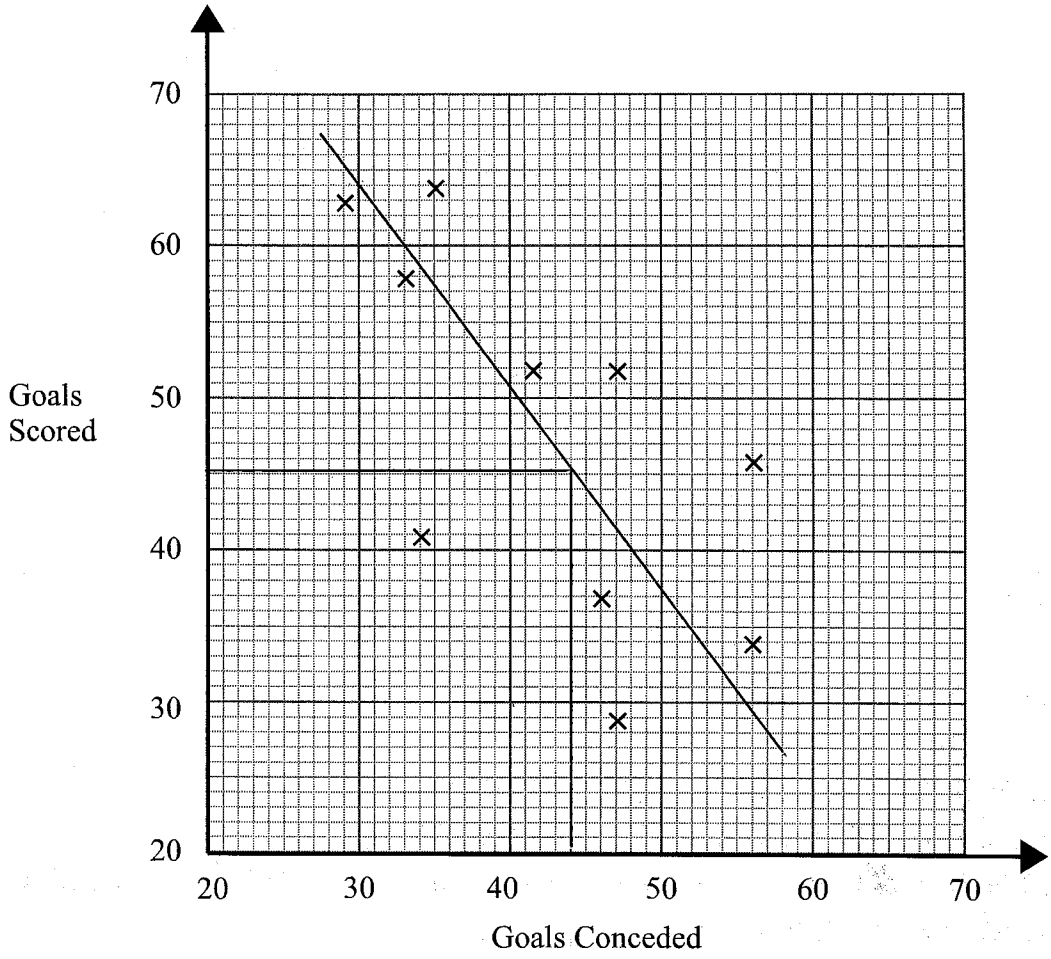
1 Gary recorded how many goals 10 football teams scored.
He also recorded how many goals they conceded

The information is shown on the scatter graph.



(a) What type of correlation does the scatter graph show?

negative
(1)

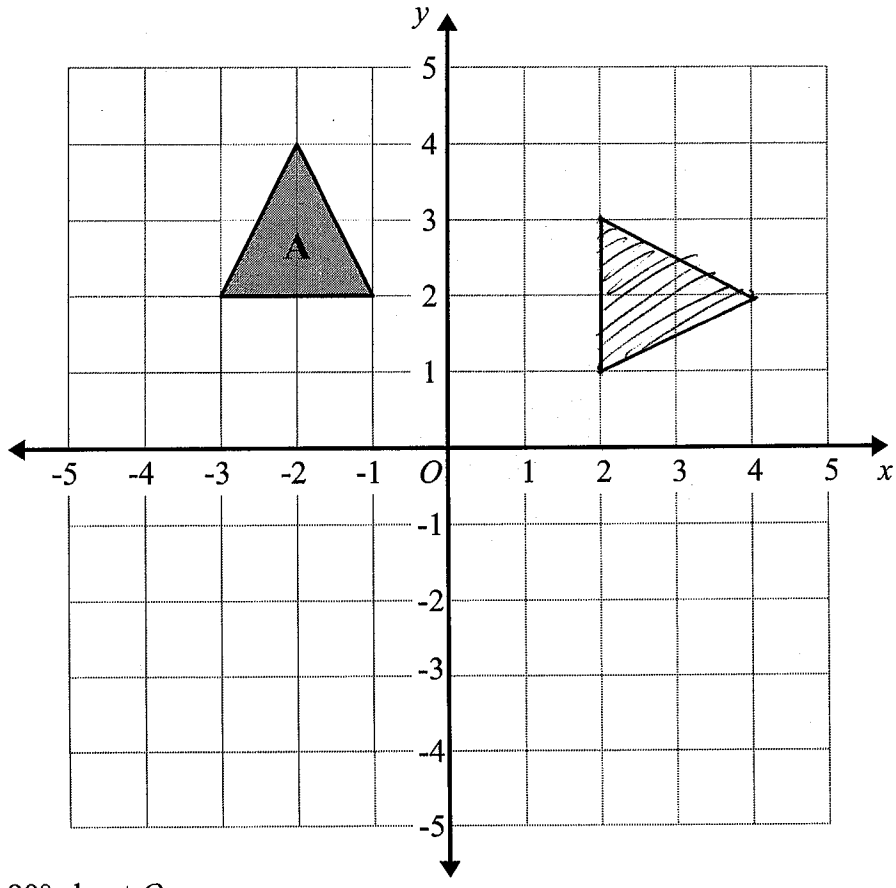
(b) Another team have conceded 44 goals.

Estimate the number of goals this team has scored.

45
(43 – 47) (2)

(Total for Question 1 is 3 marks)

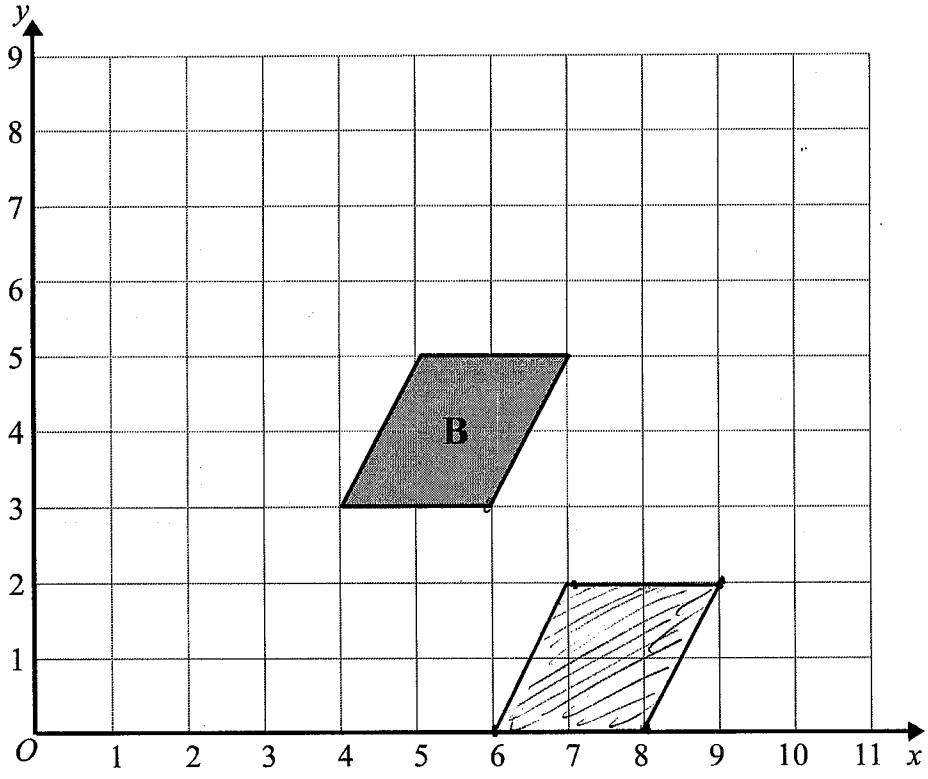
2



Rotate shape **A** 90° about O

(Total for Question 2 is 2 marks)

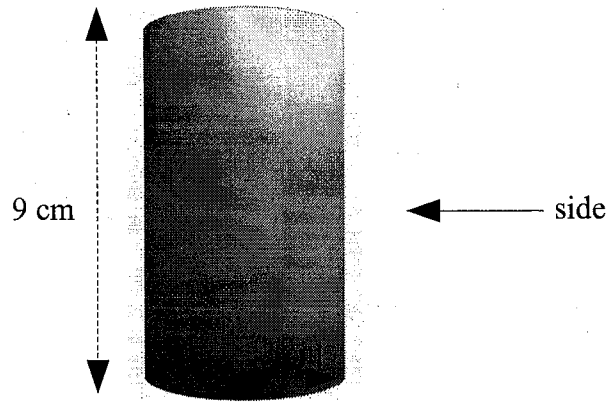
3



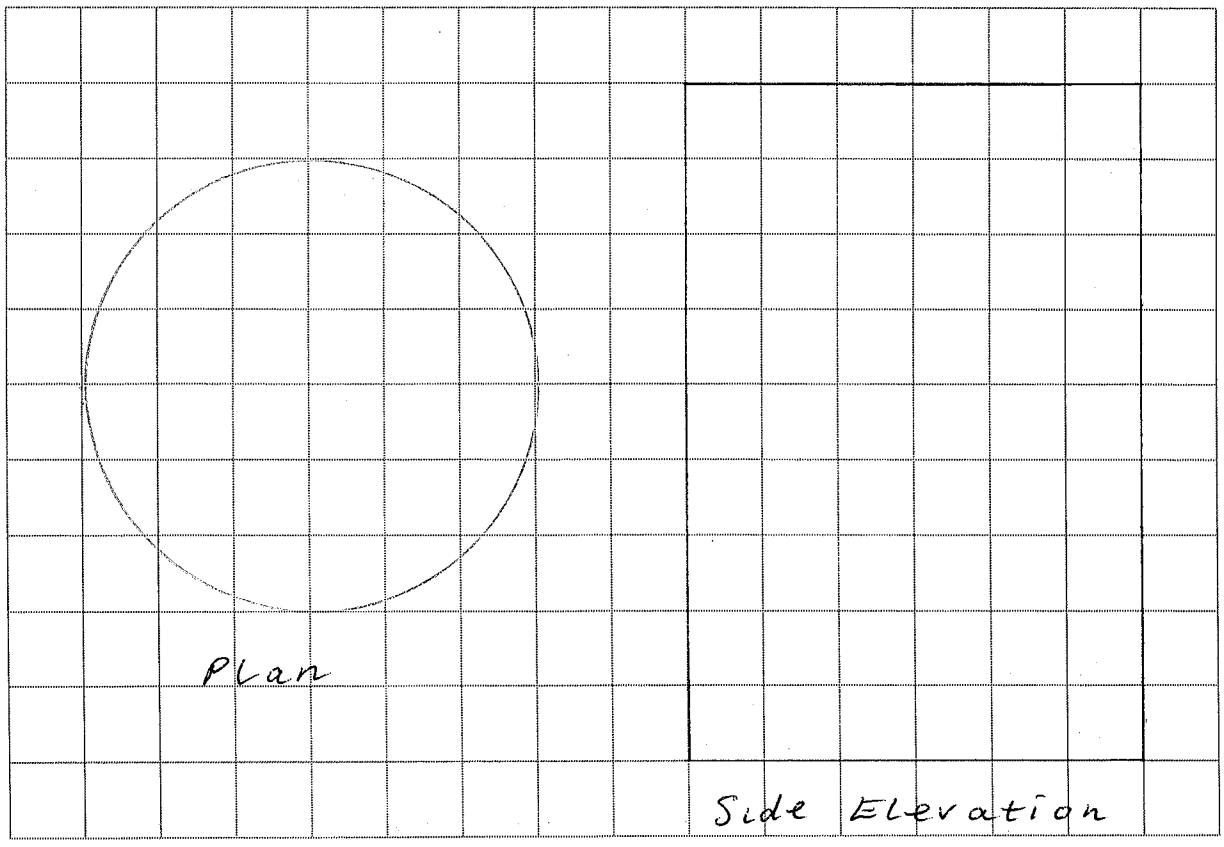
Translate shape **B** by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$

(Total for Question 3 is 2 marks)

4 The diagram shows a cylinder with radius 3 cm and perpendicular height of 9 cm

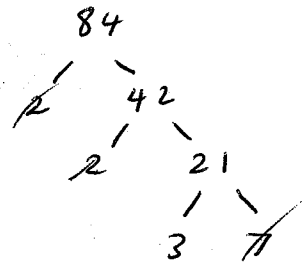


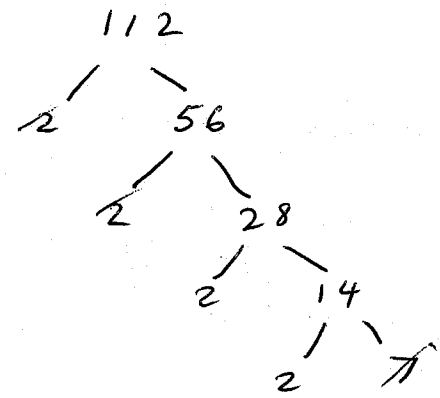
On the centimetre grid below, draw the plan and the side elevation of the cylinder.

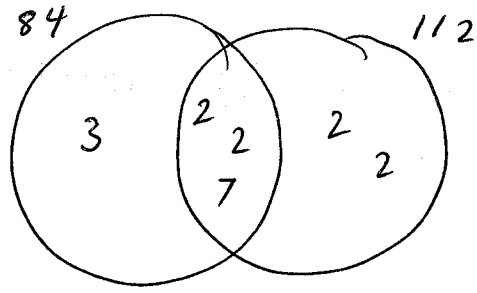


(Total for Question 4 is 3 marks)

5 Find the highest common factor (HCF) of 84 and 112







HCF = 2 × 2 × 7
= 4 × 7
= 28

28

(Total for Question 5 is 3 marks)

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6 Dennis gets paid £8.79 per hour he works.
Each week Dennis works 39 hours.

(a) Work out an estimate for the amount Dennis gets paid in a week.

9 × 40 = £360

£ 360

(2)

(b) Is your answer to part (a) an underestimate or an overestimate?
Give a reason for your answer.

both

Overestimate - I rounded the hourly pay and the number of hours up

(1)

(Total for Question 6 is 3 marks)

7 (a) Simplify $(a^4)^3$

$a^{4\times3}$

a^{12}

(1)

(b) Simplify $6m^4n^7 \div 2m^2n$

$\frac{6}{2} m^{4-2} n^{7-1}$

$3m^2n^6$

(2)

(Total for Question 7 is 3 marks)

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8 (a) Expand $6x(2y - 7)$

$12xy - 42x$

(2)

(b) Expand and Simplify $5(t - 3) + 3(2t - 1)$

$5t - 15 + 6t - 3$

$11t - 18$

(2)

(Total for Question 8 is 4 marks)

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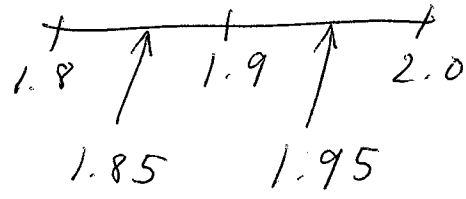
9 A number y is rounded to 1 decimal place.

The result is 1.9

Write down the error interval for y.



$1.85 \leq y < 1.95$

(Total for Question 9 is 2 marks)